is harder to address since there is so much about the microphysical basis of human thought that we've yet to understand. But much progress has been made in understanding the arrow of time when it comes to computers—undertaking, completing, and then producing a record of a computation is a basic computational sequence whose entropic properties are well understood (as developed by Charles Bennett, Rolf Landauer, and others) and fit squarely within the second law of thermodynamics. Thus, if human thought can be likened to computational processes, a similar thermodynamic explanation may apply. Notice, too, that the asymmetry associated with the fact that the universe is expanding and not contracting is related to, but logically distinct from, the arrow of time we've been exploring. If the universe's expansion were to slow down, stop, and then turn into a contraction, the arrow of time would still point in the same direction. Physical processes (eggs breaking, people aging, and so on) would still happen in the usual direction, even though the universe's expansion had reversed.

19. For the mathematically inclined reader, notice that when we make this kind of probabilistic statement we are assuming a particular probability measure: the one that is uniform over all microstates compatible with what we see right *now.* There are, of course, other measures that we could invoke. For example, David Albert in *Time and Chance* has advocated using a probability measure that is uniform over all microstates compatible with what we see *now* and what he calls *the past hypothesis*— the apparent fact that the universe began in a low-entropy state. Using this measure, we eliminate consideration of all but those histories that are compatible with the low-entropy past attested to by our memories, records, and cosmological theories. In this way of thinking, there is no probabilistic puzzle about a universe with low entropy; it began that way, by assumption, with probability 1. There is still the same huge puzzle of *why* it began that way, even if it isn't phrased in a probabilistic context.

20. You might be tempted to argue that the known universe had low entropy early on simply because it was much smaller in size than it is today, and hence—like a book with fewer pages—allowed for far fewer rearrangements of its constituents. But, by itself, this doesn't do the trick. Even a small universe can have huge entropy. For example, one possible (although unlikely) fate for our universe is that the current expansion